

AS level

Chp:- Inflation

Topic :- RPI/CPI

* NCES Questions

* Solved

* Reasons file



Zarale

Nusrat



- 1) The table gives data for an economy.

	2010	2011	2012	2013	2014
Gross Domestic Product (GDP) at current prices (\$ billion)	200	220	240	300	320
GDP deflator (price index)	100	109	125	149	154

In which year did real GDP decline compared with the previous year?

- A** 2011 **B** 2012 **C** 2013 **D** 2014

- 2) The table shows information about a country whose consumers spend their income on three commodities, P, Q and R.

commodity	index of prices in year 1	index of prices in year 2	consumers' expenditure in year 1
P	100	160	\$100 million
Q	100	80	\$300 million
R	100	100	\$100 million

Between year 1 and year 2 how has the general level of prices changed?

- A** It has risen by 40%.
B It has risen by 10%.
C It has remained the same.
D It has fallen by 5%.

- 3) Suppose the average consumer's expenditure is divided between bread, meat, milk and vegetables in the ratio 4:3:2:1.

During the course of a year, the price of bread falls by 10%, the price of meat increases by 20% and the prices of both milk and vegetables increase by 10%.

What is the increase in the average price level during the year?

- A** 5% **B** 7.3% **C** 10% **D** 12.5%

- 4) The figures in the table were taken from the consumer price indices of retail prices for three countries.

Country	2014	2015
X	100	120
Y	140	150
Z	90	99

Between 2014 and 2015, what is correct?

- A** Country X has the highest growth in the purchasing power of money.
B Country Y has the highest growth in the purchasing power of money.
C Country Z has the lowest rate of inflation.
D Country Y has the lowest rate of inflation.

- 5) The table shows the retail prices index for four countries for years 2 and 3. (Year 1 index = 100 for all countries.)

Which country had a higher rate of inflation in year 2 than year 3?

country	year 2 index	year 3 index
A	100	115
B	110	132
C	120	144
D	130	150

- 6) What is the average weighted price change illustrated by the table below?

product	percentage of income spent on product	price change %
P	10	+8
Q	15	+6
R	25	+4
S	50	-9

- A** -1.8% **B** 4.5% **C** 7.2% **D** 9.0%

- 7) The average consumer divides his expenditure between food, clothing, accommodation and transport in the ratio 2:1:4:3.

During the course of a year, the price of food falls by 2%, the price of clothing increases by 4%, the price of accommodation increases by 10% and the price of transport remains constant.

Assuming that the weights of a price index reflect the expenditure of the average consumer, what is the increase in the index over the year?

- A** 2.5% **B** 4% **C** 10% **D** 12%

- 8) According to US Consumer Prices Index (CPI) statistics, an identical good that cost \$100 in 1913 would cost \$2359.60 in 2013.

What was the approximate cumulative rate of inflation over the period?

- A** 23% **B** 230% **C** 2300% **D** 23 000%

- 9) The table shows the consumer price index (CPI) and national output at current prices in 2014 and 2015 for an economy.

year	CPI	national output at current prices
2014	100	100
2015	120	110

What can be concluded when 2015 is compared with 2014?

- A** Money national output decreased by 10%.
B Money national output increased by 20%.
C Real national output fell.
D Real national output increased.

- 10) The table shows changes in the population, price level and Gross Domestic Product (GDP) of a country.

	year 1	year 2
population (millions)	50	55
Consumer Price Index	100	120
nominal GDP (\$ billions)	400	480

What happened to real GDP and real GDP per head between year 1 and year 2?

	real GDP	real GDP per head
A	no change	fell
B	no change	rose
C	rose	fell
D	rose	rose

- 11) The data shows the consumer prices index (CPI) and an index of money wages in an economy during four years.

year	1	2	3	4
CPI	100	102	105	110
index of money wages	100	102	107	110

Which statement is **not** correct?

- A** Inflation was at its greatest between years 3 and 4.
B Real wages remained constant between years 1 and 2.
C Real wages rose between years 3 and 4.
D The rate of inflation increased over the whole period.

- 12) The table shows the levels of a country's consumer prices index (CPI) between 2015 and 2019.

	CPI
2015	100
2016	99
2017	98
2018	101
2019	102

Compared with the previous year, in which year was there disinflation but **not** deflation?

- A 2016 B 2017 C 2018 D 2019
- 13) The table shows the price indices and weights for the three commodity groups that make up the calculation of a country's consumer price index.

commodity group	index	weight
X	400	5
Y	120	3
Z	80	2

By how much has the cost of living increased since the base year?

- A 52 % B 60 % C 152 % D 520 %

- 14) The figures show the consumer price index (CPI) of a country. (1971 = 100)

	CPI
1998	200
1999	204
2000	206
2001	209

What can be deduced from the data?

- A There was hyperinflation between 1971 and 1998.
 B The rate of inflation rose in 2000.
 C The rate of inflation in 1999 was 4 %.
 D The country experienced inflation in the three years down to the end of 2001.

- 15) The table shows the share of a worker's income spent on three classes of good and the percentage change in the prices of the goods over a year.

	share of income spent on goods	change in prices of goods
clothing	$\frac{1}{4}$	+25 %
food	$\frac{1}{2}$	+50 %
fuel	$\frac{1}{4}$	-25 %

Over the same period the worker's money income rises by 25 %.

What happens to the worker's real income?

- A It falls by 25 %.
 B It falls by 50 %.
 C It rises by 25 %.
 D It is unchanged.

- 16) The table shows the weights used over three years in the UK consumer price index.

	2001	2002	2003
food	116	114	109
motoring expenditure	140	141	146
housing	205	199	203
leisure goods	49	48	48

What can be deduced from the table?

- A** Housing was a bigger proportion of the average UK household's spending in 2003 than in 2002.
- B** The average UK household ate less in 2003 than in 2001.
- C** The average UK household spent more on motoring in 2003 than in 2001.
- D** The price of leisure goods remained unchanged between 2002 and 2003.
- 17) In 2012, a survey of household expenditure identified X and Y as the contents of a basket of goods for a price index. The value of the index in 2012 was 100. In 2013, the index was valued at 116. During 2013, the prices of X and Y increased by 10% and 20%, respectively.

In percentage terms, which weights were attached to X and Y in calculating the index?

	good X (%)	good Y (%)
A	25	75
B	40	60
C	50	50
D	70	30

- 18) The table shows the price indices and weights for three commodity groups that are included in the calculation of a country's cost of living index.

commodity group	index	weight
X	300	4
Y	140	3
Z	80	3

By how much has the cost of living increased since the base year?

- A** 52% **B** 86% **C** 186% **D** 198%

$$\text{GDP deflator} = \frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100$$

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$$\text{Real GDP} = \frac{\text{Nominal GDP}}{\text{GDP deflator}}$$

- 1) The table gives data for an economy.

	2010	2011	2012	2013	2014
Gross Domestic Product (GDP) at current prices (\$ billion)	200	220	240	300	320
GDP deflator (price index)	100	109	125	149	154

In which year did real GDP decline compared with the previous year?

- A 2011 B 2012 C 2013 D 2014

- 2) The table shows information about a country whose consumers spend their income on three commodities, P, Q and R.

commodity	index of prices in year 1	index of prices in year 2	consumers' expenditure in year 1
P	100	160	\$100 million
Q	100	80	\$300 million
R	100	100	\$100 million

Between year 1 and year 2 how has the general level of prices changed?

- A It has risen by 40%.
B It has risen by 10%.
C It has remained the same.
D It has fallen by 5%.

Base year = 100

- 3) Suppose the average consumer's expenditure is divided between bread, meat, milk and vegetables in the ratio 4:3:2:1.

During the course of a year, the price of bread falls by 10%, the price of meat increases by 20% and the prices of both milk and vegetables increase by 10%.

What is the increase in the average price level during the year?

- A 5% B 7.3% C 10% D 12.5%

Bread	4	10% ↓	90	90 × 4
meat	3	20% ↑	120	120 × 3
milk	2	10% ↑	110	110 × 2
vegetables	1	10% ↑	110	110 × 1
	10			1050

* Alternative way to solve, is to use % Δ formula

$$\frac{220 - 200}{200} \times 100 = 10\%$$

$$\frac{109 - 100}{100} \times 100 = 9\%$$

$$\text{Real} = 10\% - 9\% = 1\%$$

$$100 \times 100 = 10000$$

$$100 \times 300 = 30000$$

$$100 \times 100 = 10000$$

1

2

$$160 \times 100 = 16000$$

$$80 \times 300 = 24000$$

$$100 \times 100 = 10000$$

50K

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- 4) The figures in the table were taken from the consumer price indices of retail prices for three countries.

Country	2014	2015
X	100	120
Y	140	150
Z	90	99

$$\% \Delta = \frac{\text{New} - \text{old}}{\text{old}} \times 100$$

$$20\% \text{ for X}$$

$$7.14\% \text{ for Y}$$

$$10\% \text{ for Z}$$

Between 2014 and 2015, what is correct?

- A Country X has the highest growth in the purchasing power of money.
B Country Y has the highest growth in the purchasing power of money.
C Country Z has the lowest rate of inflation.
D Country Y has the lowest rate of inflation.

Data shows rate of inflation

- 5) The table shows the retail prices index for four countries for years 2 and 3. (Year 1 index = 100 for all countries.)

country	year 2 index	year 3 index
A	100	115
B	100	110
C	100	120
D	100	130

Which country had a higher rate of inflation in year 2 than year 3?

$$\% \Delta = \frac{\text{New} - \text{old}}{\text{old}} \times 100$$

year 2 > year 3

- 6) What is the average weighted price change illustrated by the table below?

product	percentage of income spent on product	price change %
P	10	+8
Q	15	+6
R	25	+4
S	50	-9

$$-1.8\%$$

$$98.2 - 100 = -1.8\%$$

$$108 \times 10 = 1080$$

$$106 \times 15 = 1590$$

$$104 \times 25 = 2600$$

$$91 \times 50 = 4550$$

$$\frac{9820}{100} = 98.2$$

- B 4.5% C 7.2% D 9.0%

Food	2	2% ↓ 98
Clothing	1	4% ↑ 104
Accommodation	4	10% ↑ 110
transport	3	- 100
	<u>10</u>	

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98 × 2
104 × 1
110 × 4
100 × 3
<u>1040</u>

$$\frac{1040}{10} = 104 - 100 = 4\%$$

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- 7) The average consumer divides his expenditure between food, clothing, accommodation and transport in the ratio 2:1:4:3.

During the course of a year, the price of food falls by 2%, the price of clothing increases by 4%, the price of accommodation increases by 10% and the price of transport remains constant.

Assuming that the weights of a price index reflect the expenditure of the average consumer, what is the increase in the index over the year?

- A 2.5% ☒ B 4% C 10% D 12%

- 8) According to US Consumer Prices Index (CPI) statistics, an identical good that cost \$100 in 1913 would cost \$2359.60 in 2013.

What was the approximate cumulative rate of inflation over the period?

- A 23% B 230% ☒ C 2300% D 23000%

$$\% \Delta = \frac{\text{New} - \text{old}}{\text{old}} \times 100$$

- 9) The table shows the consumer price index (CPI) and national output at current prices in 2014 and 2015 for an economy.

year	CPI	national output at current prices
2014	100	100
2015	120	110

Nominal

20% ↑

10% ↑

What can be concluded when 2015 is compared with 2014?

- A Money national output decreased by 10%. increased
 B Money national output increased by 20%.
☒ C Real national output fell. Nominal - rate of inflation, 10% - 20% = -10%.
 D Real national output increased.

- 10) The table shows changes in the population, price level and Gross Domestic Product (GDP) of a country.

	year 1	year 2
population (millions)	50	55
Consumer Price Index	100	120
nominal GDP (\$ billions)	400	480

10%.

20%.

20%.

} Real (no change) GDP

What happened to real GDP and real GDP per head between year 1 and year 2?

	real GDP	real GDP per head
☒ A	no change	fell
B	no change	rose
C	rose	fell
D	rose	rose

Nominal - deflation

→ then divide by population

Real GDP / population, =, Real GDP per head ↓

- 11) The data shows the consumer prices index (CPI) and an index of money wages in an economy during four years.

year	1	2	3	4
CPI	100	102	105	110
index of money wages	100	102	107	110

2%.

2.9%.

4.8%.

2%.

4.9%.

2.8%.

Which statement is not correct?

- A Inflation was at its greatest between years 3 and 4. yes, 4.8%.
 B Real wages remained constant between years 1 and 2. yes, 2% - 2% = 0%.
☒ C Real wages rose between years 3 and 4. it fell, 2.8 - 4.8 = -2%.
 D The rate of inflation increased over the whole period. yes, 2%, 2.9%, 4.8%.

- 12) The table shows the levels of a country's consumer prices index (CPI) between 2015 and 2019.

	CPI
2015	100
2016	99
2017	98
2018	101
2019	102

-1%
 -1.01%
 3.06%
 0.99%
 Rate of inflation ↓
 negative inflation

Compared with the previous year, in which year was there disinflation but not deflation?

- A 2016 B 2017 C 2018 ☒ D 2019

- 13) The table shows the price indices and weights for the three commodity groups that make up the calculation of a country's consumer price index.

commodity group	index	weight
X	400	5
Y	120	3
Z	80	2

$$\frac{2520}{10} = 252 - 100 = 152\%$$

By how much has the cost of living increased since the base year?

- A 52% B 60% ☒ C 152% D 520%

- 14) The figures show the consumer price index (CPI) of a country. (1971 = 100)

	CPI
1998	200
1999	204
2000	206
2001	209

2%
 0.98%
 1.45%

What can be deduced from the data?

- A There was hyperinflation between 1971 and 1998. *NO, 2% rate of inflation*
 B The rate of inflation rose in 2000. *rate ↓, 2% to 0.98%*
 C The rate of inflation in 1999 was 4%. *NO, it was 2%*
☒ D The country experienced inflation in the three years down to the end of 2001. *Yes, all positive*

- 15) The table shows the share of a worker's income spent on three classes of good and the percentage change in the prices of the goods over a year.

	share of income spent on goods	change in prices of goods
clothing	$\frac{1}{4}$	+25%
food	$\frac{1}{2}$	+50%
fuel	$\frac{1}{4}$	-25%

125
 150
 75

Over the same period the worker's money income rises by 25%.

What happens to the worker's real income?

- A It falls by 25%.
 B It falls by 50%.
 C It rises by 25%.
☒ D It is unchanged.

$125 - 100 = 25$ (rate of inflation)
 Money income ↑ 25%
 Rate of inflation ↑ 2%
 Real income = unchanged

→ shows proportion of income

- 16) The table shows the weights used over three years in the UK consumer price index.

	2001	2002	2003
food	116	114	109
motoring expenditure	140	141	146
housing	205	199	203
leisure goods	49	48	48

What can be deduced from the table?

- ✓ **A** Housing was a bigger proportion of the average UK household's spending in 2003 than in 2002. *See what is meant by 'weights'*
- B** The average UK household ate less in 2003 than in 2001.
- C** The average UK household spent more on motoring in 2003 than in 2001.
- D** The price of leisure goods remained unchanged between 2002 and 2003.

- 17) In 2012, a survey of household expenditure identified X and Y as the contents of a basket of goods for a price index. The value of the index in 2012 was 100. In 2013, the index was valued at 116. During 2013, the prices of X and Y increased by 10% and 20%, respectively.

16%
* In percentage terms, which weights were attached to X and Y in calculating the index?

	good X (%)	good Y (%)
A	25	75
✓ B	40	60
C	50	50
D	70	30

$$\begin{aligned}
 10\% \uparrow & 110 \times 40 = 4400 \\
 20\% \uparrow & 120 \times 60 = 7200 \\
 & \underline{100} \quad \underline{11600}
 \end{aligned}$$

$$\frac{11600}{100} = 116$$

- 18) The table shows the price indices and weights for three commodity groups that are included in the calculation of a country's cost of living index.

commodity group	index	weight
X	300	4
Y	140	3
Z	80	3

$$\frac{1860}{10} = 186 - 100 = 86\%$$

By how much has the cost of living increased since the base year?

- A** 52% ✓ **B** 86% **C** 186% **D** 198%

INFLATION: PRICE INDICIES

AS LEVEL

ECONOMICS P1

1.	B	7.	B	13.	C
2.	C	8.	C	14.	D
3.	A	9.	C	15.	D
4.	D	10.	A	16.	A
5.	D	11.	C	17.	B
6.	A	12.	D	18.	B

**INFLATION:
PRICE INDICIES
(CALCULATIONS)**

1) The table gives data for an economy.

	2010	2011	2012	2013	2014
Gross Domestic Product (GDP) at current prices (\$ billion)	200	220	240	300	320
GDP deflator (price index)	100	109	125	149	154

In which year did real GDP decline compared with the previous year?

- A** 2011 **B** 2012 **C** 2013 **D** 2014

1. ANSWER: B

In 2011, GDP rose by 10% $\left(\frac{(220-200)}{200} \times 100\right)$, whilst the GDP deflator rose by 9% $\left(\frac{(109-100)}{100} \times 100\right)$, this shows that there was rise in real GDP, as the increase in nominal GDP is greater than the rise in prices. In 2012 the rise in GDP was of 9.1% $\left(\frac{(240-220)}{220} \times 100\right)$, while the rise in the GDP deflator was of 14.7% $\left(\frac{(125-109)}{109} \times 100\right)$, as the change in prices was greater than the change in nominal GDP, thus real GDP should have fallen.

- 2) The table shows information about a country whose consumers spend their income on three commodities, P, Q and R.

commodity	index of prices in year 1	index of prices in year 2	consumers' expenditure in year 1
P	100	160	\$100 million
Q	100	80	\$300 million
R	100	100	\$100 million

Between year 1 and year 2 how has the general level of prices changed?

- A It has risen by 40%.
- B It has risen by 10%.
- C It has remained the same.
- D It has fallen by 5%.

2. ANSWER: C

The percentage change in the price of commodities P, Q, and R, are of 60%, -20%, and 0% respectively. Furthermore, the weight assigned to each good can also be calculated by expressing the expenditure on the commodity to the total consumers' expenditure, this gives

commodity P's weight as $\frac{1}{5} \left(\frac{100m}{500m} \right)$, commodity Q's becomes $\frac{3}{5} \left(\frac{300m}{500m} \right)$,

and commodity R's is $\frac{1}{5} \left(\frac{100m}{500m} \right)$. Using this the weighted average price

change can be found by multiplying the price change, with the weights,

thus the this becomes $0 \left(\left(6 \times \frac{1}{5} \right) + \left(-2 \times \frac{3}{5} \right) + \left(0 \times \frac{1}{5} \right) = \frac{6}{5} - \frac{6}{5} \right)$,

therefore there was no change in the price level between year 1, and year 2.

- 3) Suppose the average consumer's expenditure is divided between bread, meat, milk and vegetables in the ratio 4:3:2:1.

During the course of a year, the price of bread falls by 10%, the price of meat increases by 20% and the prices of both milk and vegetables increase by 10%.

What is the increase in the average price level during the year?

- A** 5% **B** 7.3% **C** 10% **D** 12.5%

3. ANSWER: A

The average price level is calculated, by multiplying the price change by the weight/percentage of income spent on the product. Thus, the

average price level is $5\% \left(\left(-10 \times \frac{4}{10} \right) + \left(20 \times \frac{3}{10} \right) + \left(10 \times \frac{2}{10} \right) + \left(10 \times \frac{1}{10} \right) \right)$.

- 4) The figures in the table were taken from the consumer price indices of retail prices for three countries.

Country	2014	2015
X	100	120
Y	140	150
Z	90	99

Between 2014 and 2015, what is correct?

- A Country X has the highest growth in the purchasing power of money.
- B Country Y has the highest growth in the purchasing power of money.
- C Country Z has the lowest rate of inflation.
- D Country Y has the lowest rate of inflation.

4. ANSWER: D

The rate of inflation may be found by finding the change in the consumer price index between the two years, thus the rate of inflation for Country Y, is calculated as $7.14\% \left(\frac{(150-140)}{140} \times 100 \right)$. This is the lowest rate of inflation as compared to the remaining two countries.

- 5) The table shows the retail prices index for four countries for years 2 and 3.
(Year 1 index = 100 for all countries.)

Which country had a higher rate of inflation in year 2 than year 3?

country	year 2 index	year 3 index
A	100	115
B	110	132
C	120	144
D	130	150

5. ANSWER: D

In year 2, there was a 30% $\left(\frac{(130-100)}{100} \times 100\right)$ rate of inflation, while in year 3, the rate of inflation was of 15.4% $\left(\frac{(150-130)}{130} \times 100\right)$. Thus, the rate of inflation was lower in year 3, than in year 2.

- 6) What is the average weighted price change illustrated by the table below?

product	percentage of income spent on product	price change %
P	10	+8
Q	15	+6
R	25	+4
S	50	-9

- A** -1.8% **B** 4.5% **C** 7.2% **D** 9.0%

6. ANSWER: D

The average weighted price change is calculated, by multiplying the price change by the weight/percentage of income spent on the product. Thus, the average weighted index is -1.8% $((0.08 \times 10) + (0.06 \times 15) + (0.04 \times 25) + (-0.09 \times 0.5))$.

- 7) The average consumer divides his expenditure between food, clothing, accommodation and transport in the ratio 2 : 1 : 4 : 3.

During the course of a year, the price of food falls by 2%, the price of clothing increases by 4%, the price of accommodation increases by 10% and the price of transport remains constant.

Assuming that the weights of a price index reflect the expenditure of the average consumer, what is the increase in the index over the year?

- A** 2.5% **B** 4% **C** 10% **D** 12%

7. ANSWER: B

The average weighted price change is calculated, by multiplying the price change by the weight/percentage of income spent on the product. Thus, the average weighted index is 4% $\left(\left(-2 \times \frac{2}{10} \right) + \left(4 \times \frac{1}{10} \right) + \left(10 \times \frac{4}{10} \right) + \left(0 \times \frac{3}{10} \right) \right)$.

- 8) According to US Consumer Prices Index (CPI) statistics, an identical good that cost \$100 in 1913 would cost \$2359.60 in 2013.

What was the approximate cumulative rate of inflation over the period?

- A** 23% **B** 230% **C** 2300% **D** 23000%

8. ANSWER: C

The rate of inflation between 1913, and 2013 is calculated as 2259.3% $\left(\frac{(2359.6-100)}{100} \times 100 \right)$, which approximates to about 2300%.

- 9) The table shows the consumer price index (CPI) and national output at current prices in 2014 and 2015 for an economy.

year	CPI	national output at current prices
2014	100	100
2015	120	110

What can be concluded when 2015 is compared with 2014?

- A** Money national output decreased by 10%.
B Money national output increased by 20%.
C Real national output fell.
D Real national output increased.

9. ANSWER: C

The change in the price level was of 20%, whereas the change in the nominal GDP, or the GDP at current prices, was of 10%. Therefore the impact on the real GDP was of -10% (10 – 20), thus, the real national output / real GDP fell.

- 10) The table shows changes in the population, price level and Gross Domestic Product (GDP) of a country.

	year 1	year 2
population (millions)	50	55
Consumer Price Index	100	120
nominal GDP (\$ billions)	400	480

What happened to real GDP and real GDP per head between year 1 and year 2?

	real GDP	real GDP per head
A	no change	fell
B	no change	rose
C	rose	fell
D	rose	rose

10. ANSWER: A

It can be seen that there was a 20% $\left(\frac{(120-100)}{100} \times 100\right)$ increase in the price level between year 1, and 2. To calculate real GDP this increase in price level must be excluded from the nominal GDP figure of year 2, to do this the figure should be divided by 120%, which gives the value of real GDP as \$400bn $\left(\frac{480}{120\%}\right)$, which is the same as year 1, so there was no change in real GDP. On the other hand, the real GDP per head in year 1 was \$8000 $\left(\frac{400,000}{50}\right)$, while in year 2 it was \$7273 $\left(\frac{400,000}{55}\right)$, thus the real GDP per head fell between year 1, and year 2.

- 11) The data shows the consumer prices index (CPI) and an index of money wages in an economy during four years.

year	1	2	3	4
CPI	100	102	105	110
index of money wages	100	102	107	110

Which statement is **not** correct?

- A Inflation was at its greatest between years 3 and 4.
- B Real wages remained constant between years 1 and 2.
- C Real wages rose between years 3 and 4.
- D The rate of inflation increased over the whole period.

11. ANSWER: C

The table shows that in year 2 both CPI and the index of money wages, increased by the same amount, hence there should not have been a change in the real wages, between year 1, and year 2, therefore option B is correct. Similarly, as the CPI was highest in years 3, and 4, therefore inflation would have been the highest between these years, hence option A is also correct. Furthermore, if the rate of inflation is calculated the it can also be seen that each year the rate of inflation was higher than the previous year, therefore the rate of inflation increased throughout the period. Thus, option C is incorrect, because the index at year 4, stood at the same amount, thus the real wages would have remained constant.

12) The table shows the levels of a country's consumer prices index (CPI) between 2015 and 2019.

	CPI
2015	100
2016	99
2017	98
2018	101
2019	102

Compared with the previous year, in which year was there disinflation but **not** deflation?

- A** 2016 **B** 2017 **C** 2018 **D** 2019

12. ANSWER: D

The decrease in the price level from 2015 to 2016 is of -1%. As the rate of inflation is negative therefore there is deflation in the economy. In 2017, the CPI fell to 98, from 99, again there was a -1% change in the price level, therefore the economy also experienced deflation in this year. In 2018 the economy's CPI rose to 101, therefore there was inflation of 3%. In 2019 the CPI was 102, there was an inflation rate of 0.99%, therefore as the rate of inflation is lower than the previous year, the economy experienced disinflation.

- 13) The table shows the price indices and weights for the three commodity groups that make up the calculation of a country's consumer price index.

commodity group	index	weight
X	400	5
Y	120	3
Z	80	2

By how much has the cost of living increased since the base year?

- A** 52 % **B** 60 % **C** 152 % **D** 520 %

13. ANSWER: C

In the base year the weight is 100, therefore for the change in prices we must compare this value with those given in the index column.

Thus, the increase in price of commodity X is of 300%. While for commodity Y, and Z it is 20%, and -20%, respectively. In order to find the weighted average price level the weight must be multiplied with the price change, therefore the average weighted price level is calculated

as $152\% \left(\left(300 \times \frac{5}{10} \right) + \left(20 \times \frac{3}{10} \right) + \left(-20 \times \frac{2}{10} \right) \right)$.

14) The figures show the consumer price index (CPI) of a country. (1971 = 100)

	CPI
1998	200
1999	204
2000	206
2001	209

What can be deduced from the data?

- A There was hyperinflation between 1971 and 1998.
- B The rate of inflation rose in 2000.
- C The rate of inflation in 1999 was 4 %.
- D The country experienced inflation in the three years down to the end of 2001.

14. ANSWER: D

As the country experienced a rising inflation rate, and CPI in each subsequent year, therefore the economy experienced inflation in the three year down to 2001.

- 15) The table shows the share of a worker's income spent on three classes of good and the percentage change in the prices of the goods over a year.

	share of income spent on goods	change in prices of goods
clothing	$\frac{1}{4}$	+25 %
food	$\frac{1}{2}$	+50 %
fuel	$\frac{1}{4}$	-25 %

Over the same period the worker's money income rises by 25 %.

What happens to the worker's real income?

- A It falls by 25 %.
- B It falls by 50 %.
- C It rises by 25 %.
- D It is unchanged.

15. ANSWER: D

The money income increase by 25%. In order to know about the real income, we must remove the effect of the rise in prices, therefore the average weighted price change must be calculated. In order to find the weighted average price level the weight must be multiplied with the price change, therefore the average weighted price level is calculated as $25\% \left(\left(25 \times \frac{1}{4} \right) + \left(50 \times \frac{1}{2} \right) + \left(-25 \times \frac{1}{4} \right) \right)$. As both the price level and money wages increase by 25% therefore the real income of the worker remains unchanged (25 – 25).

- 16) The table shows the weights used over three years in the UK consumer price index.

	2001	2002	2003
food	116	114	109
motoring expenditure	140	141	146
housing	205	199	203
leisure goods	49	48	48

What can be deduced from the table?

- A Housing was a bigger proportion of the average UK household's spending in 2003 than in 2002.
- B The average UK household ate less in 2003 than in 2001.
- C The average UK household spent more on motoring in 2003 than in 2001.
- D The price of leisure goods remained unchanged between 2002 and 2003.

16. ANSWER: A

Weights are the proportion of expenditure of the household as compared to the total income of the household. Therefore, they do not indicate the change in prices of commodities, thus option D is incorrect. Moreover, as weights indicate a proportion of income, thus any change in the income of the household will cause the weight of the household to change. Thus it can also not be determined whether a household **spent more, or less** on one commodity in a year as compared to another, it is more accurate to say that the household spent a higher / lower proportion of income on that commodity. Therefore, option A is correct.

- 17) In 2012, a survey of household expenditure identified X and Y as the contents of a basket of goods for a price index. The value of the index in 2012 was 100. In 2013, the index was valued at 116. During 2013, the prices of X and Y increased by 10% and 20%, respectively.

In percentage terms, which weights were attached to X and Y in calculating the index?

	good X (%)	good Y (%)
A	25	75
B	40	60
C	50	50
D	70	30

17. ANSWER: B

In 2013, the index was valued at 116, this shows that the change in the price level would have been of 16%, as in 2012, the index was at 100. Therefore, the weighted price index must equal to 16%. With the change in the price of good X, and Y being 10%, and 20%, respectively, as figure of the weighted price index is calculated when the weight of good X, is 40%, and of Y, is 60% $((10 \times 0.4) + (20 \times 0.6) = 4 + 12 = 16)$.

- 18) The table shows the price indices and weights for three commodity groups that are included in the calculation of a country's cost of living index.

commodity group	index	weight
X	300	4
Y	140	3
Z	80	3

By how much has the cost of living increased since the base year?

- A** 52 % **B** 86 % **C** 186 % **D** 198 %

18. ANSWER: C

In the base year the weight is 100, therefore for the change in prices we must compare this value with those given in the index column.

Thus, the increase in price of commodity X is of 200%. While for commodity Y, and Z it is 40%, and -20%, respectively. In order to find the weighted average price level the weight must be multiplied with the price change, therefore the average weighted price level is calculated

as $86\% \left(\left(200 \times \frac{4}{10} \right) + \left(40 \times \frac{3}{10} \right) + \left(-20 \times \frac{3}{10} \right) \right)$.